CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

Project Title: Customer Churn Prediction Using Machine Learning

Submitted By

Name: Nayna Ughade

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DECLARATION

I, Nayna Ughade, hereby declare that the project titled "Customer Churn Prediction Using Machine Learning" is my original work carried out as part of my MBA curriculum. The work presented in this report has not been submitted previously for any degree or diploma. The information and results presented are based on my learning and implementation using publicly available datasets.

Date: _____

Signature: __

ACKNOWLEDGEMENT

I express my sincere gratitude to my college for providing the opportunity to work on this project. I also thank the online learning resources and open-source communities that helped me understand machine learning concepts and tools. Finally, I thank my family and friends for encouraging and supporting me throughout the completion of this project.

ABSTRACT

Customer retention has become one of the most important objectives for companies operating in competitive industries. Acquiring a new customer is more expensive than retaining an existing one. Therefore, predicting customer churn helps businesses identify customers who are likely to discontinue their services.
This project uses the Telco Customer Churn dataset to build a machine learning model capable of predicting customer churn. The project includes data preprocessing, exploratory data analysis, feature encoding, model training using Logistic Regression, and performance evaluation. The developed model achieved an accuracy of 82.04%, demonstrating that machine learning can effectively support customer retention strategies.

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INTRODUCTION

Customer churn refers to the percentage of customers who stop using a company's product or service over a specific period. High customer churn negatively impacts business growth, profitability, and customer lifetime value.

Machine Learning enables organizations to analyze historical customer behavior and predict customers who are at risk of leaving. These predictions allow businesses to implement targeted retention strategies, improve customer satisfaction, and reduce revenue loss.

INDUSTRY OVERVIEW

The telecommunications industry is highly competitive, with customers having multiple service providers to choose from. Companies continuously invest in customer acquisition and retention. Predictive analytics has become an essential business tool for understanding customer behavior and reducing churn.

PROBLEM STATEMENT

Telecommunication companies lose revenue when customers discontinue their services. Identifying customers who are likely to churn is difficult using traditional analysis methods. Therefore, an intelligent prediction model is required to identify such customers before they leave.

OBJECTIVES

- To study customer churn behaviour.
- To analyze telecom customer data.
- To preprocess customer data for machine learning.
- To build a Logistic Regression prediction model.
- To evaluate model performance.
- To provide business recommendations for reducing churn.

SCOPE OF THE PROJECT

The project focuses on predicting customer churn using historical telecom customer information. It demonstrates how machine learning techniques can support business decision-making. The project can be extended to other industries such as banking, insurance, healthcare, and e-commerce.

DATASET DESCRIPTION

The project uses the Telco Customer Churn Dataset.

Dataset Details:

- Total Customers: 7043
- Total Variables: 21
- Target Variable: Churn

Key Features include:

- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Internet Service
- Contract
- Payment Method
- Monthly Charges
- Total Charges

TOOLS AND TECHNOLOGIES

- Software: Google Colab
- Programming Language: Python
- Libraries: Pandas

  NumPy

  Matplotlib

  Seaborn

  Scikit-learn

METHODOLOGY

The project followed the CRISP-DM methodology.

Step 1 - Data Collection

Step 2 - Data Cleaning

Step 3 - Exploratory Data Analysis

Step 4 - Feature Encoding

Step 5 - Data Splitting

Step 6 - Model Training

Step 7 - Prediction

Step 8 - Performance Evaluation

DATA PREPROCESSING

The preprocessing stage involved:

- Importing the dataset
- Removing missing values
- Converting TotalCharges into numeric format
- Removing customerID
- Applying Label Encoding
- Splitting data into training and testing datasets

EXPLORATORY DATA ANALYSIS

The following visualizations were created:

- Customer Churn Distribution
- Gender vs Churn
- Contract Type vs Churn
- Monthly Charges vs Churn
- Tenure vs Churn
- Internet Service vs Churn

Key Findings

- Customers with month-to-month contracts showed the highest churn.
- Higher monthly charges were associated with increased churn.
- Customers with longer tenure were less likely to churn.

(Insert your graphs here.)

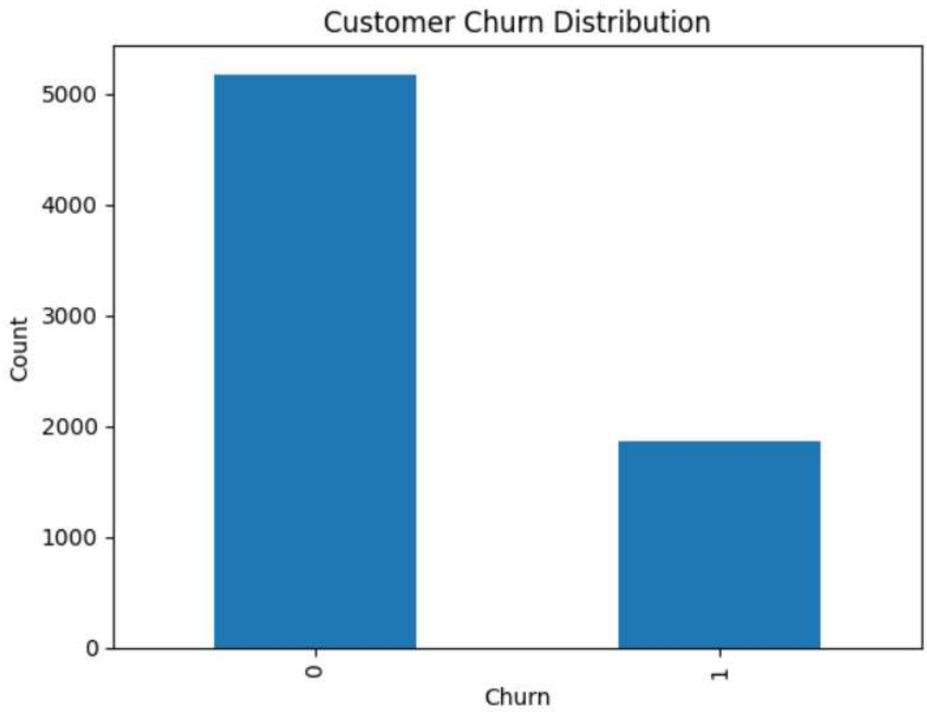
MODEL DEVELOPMENT

Algorithm Used:
Logistic Regression

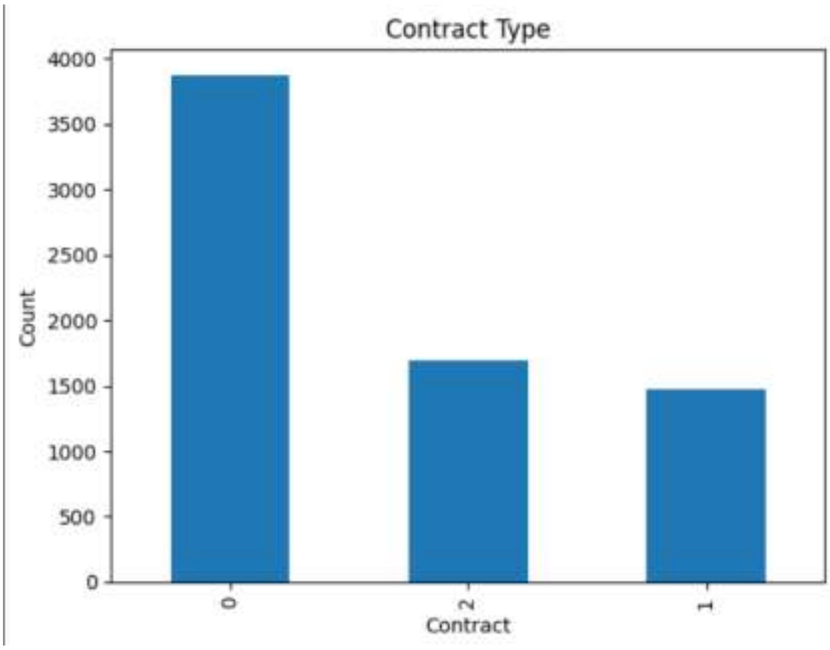
The model was trained using an 80:20 train-test split.

Performance metrics:

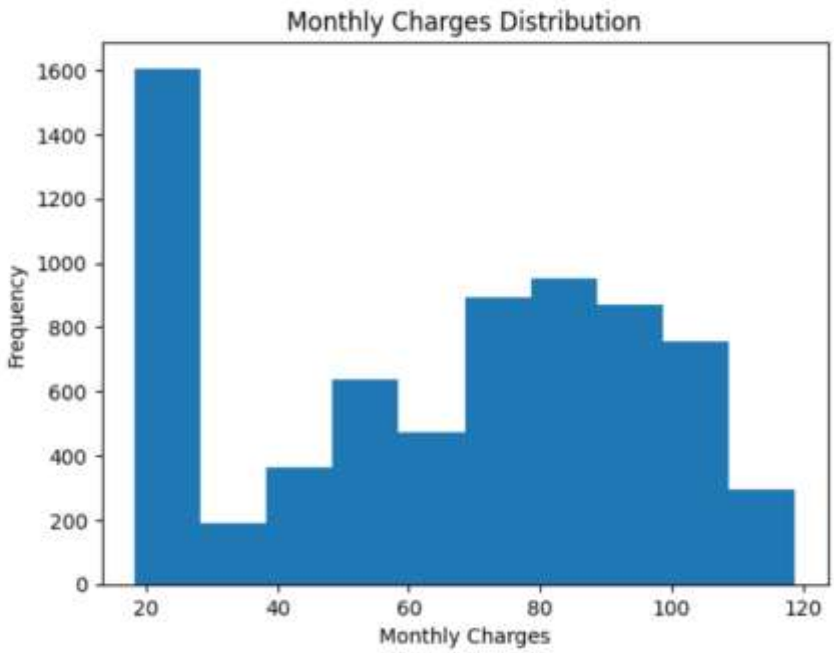
- Accuracy
- Confusion Matrix
- Classification Report



Description: This graph shows the number of customers who stayed and who left the company. It helps understand the class distribution in the dataset.



Description: This graph shows the number of customers under each contract type. Customers with month-to-month contracts generally have a higher chance of churning.



Description: This graph shows how monthly charges are distributed among customers. It helps identify whether customers with higher monthly charges are more likely to churn.

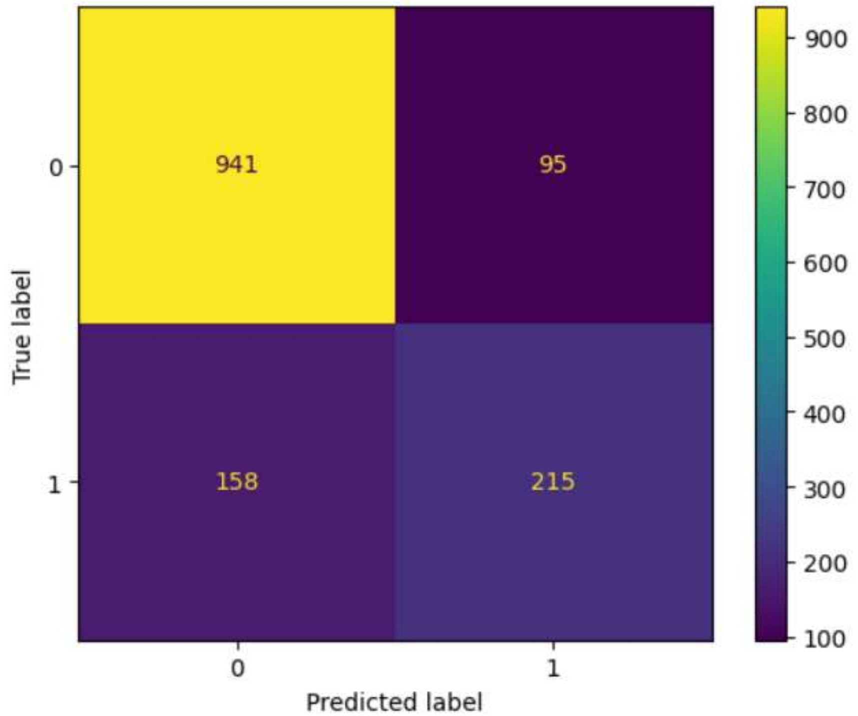
RESULTS AND DISCUSSION

Accuracy : 82.04%

Accuracy: The Logistic Regression model achieved an accuracy of 82.04%, indicating that it correctly predicted customer churn for approximately 82 out of every 100 customers.

Confusion Matrix :

[[941 95]
 [158 215]]



Confusion Matrix: The confusion matrix shows that the model correctly classified most customer records, while only a limited number of predictions were incorrect.

Interpretation

The model correctly classified most customers and achieved an overall accuracy of 82.04%. It performed well in identifying customers who remained with the company while providing useful predictions for customers at risk of churn.

(Insert screenshots of your Colab outputs here.)

BUSINESS INSIGHTS

The analysis provides the following insights:

- Month-to-month contract customers are more likely to churn.
- Customers paying higher monthly charges have a higher probability of leaving.
- Long-term customers are generally more loyal.
- Businesses can reduce churn by offering long-term plans, loyalty rewards, and personalized offers.

CONCLUSION

This project demonstrates how machine learning can be used to solve a real-world business problem. The Logistic Regression model achieved 82.04% accuracy in predicting customer churn. The findings can help telecom companies identify high-risk customers and improve customer retention strategies.

LIMITATIONS

- Only one dataset was used.
- Only Logistic Regression was implemented.
- The model may perform differently on data from other companies.
- Additional features may improve prediction accuracy.

FUTURE SCOPE

- Compare multiple machine learning algorithms.
- Develop a real-time prediction dashboard.
- Deploy the model as a web application.
- Use deep learning techniques for improved prediction.

REFERENCES

1. IBM Telco Customer Churn Dataset
2. Python Documentation
3. Pandas Documentation
4. Scikit-learn Documentation
5. Google Colab Documentation